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# Practical 5: Ridge and Lasso Regression

# 1. Import libraries
import numpy as np
import pandas as pd

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge, Lasso
from sklearn.metrics import mean_squared_error, r2_score

# 2. Load the dataset
diabetes = load_diabetes()

X = diabetes.data
y = diabetes.target

# 3. Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 4. Scale the features
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# 5. Create Ridge and Lasso models
ridge_model = Ridge(alpha=1.0)
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lasso_model = Lasso(alpha=0.1, max_iter=10000)

# 6. Train the models
ridge_model.fit(X_train, y_train)
lasso_model.fit(X_train, y_train)

# 7. Make predictions
ridge_pred = ridge_model.predict(X_test)
lasso_pred = lasso_model.predict(X_test)

# 8. Evaluate Ridge Regression
print("RIDGE REGRESSION")
print("MSE:", mean_squared_error(y_test, ridge_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, ridge_pred)))
print("R2 Score:", r2_score(y_test, ridge_pred))

# 9. Evaluate Lasso Regression
print("\nLASSO REGRESSION")
print("MSE:", mean_squared_error(y_test, lasso_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, lasso_pred)))
print("R2 Score:", r2_score(y_test, lasso_pred))

# 10. Compare the models
comparison = pd.DataFrame({
    "Model": ["Ridge", "Lasso"],
    "MSE": [
        mean_squared_error(y_test, ridge_pred),
        mean_squared_error(y_test, lasso_pred)
    ],
    "R2 Score": [
        r2_score(y_test, ridge_pred),
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        r2_score(y_test, lasso_pred)
    ]
})

print("\nModel Comparison:")
print(comparison)
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